The chain

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Chapter 08 Stage 8 of 14 Extreme concentration · Hard to substitute

Foundries and Fabs

The newest chips need the know-how to build the newest transistors and a fab that costs tens of billions of dollars. Only TSMC, Intel and Samsung have both.

Specimen 08 · Foundries and Fabs

A gate-all-around transistor, the switch in a 2 nm-class chip, drawn at true proportions. The three silicon sheets are each 6 nm thick and 30 nm wide. Each sheet is a few dozen atoms from top to bottom, roughly ten thousand times thinner than a human hair. The gold cage is the gate metal wrapping all four sides of each sheet.

In plain terms

A chip factory, or fab, builds the transistors, the billions of tiny switches in every chip, and then wires them together. Each switch works like a valve. Current runs along a narrow strip of silicon called the channel, and a small voltage on a gate above the strip opens or shuts the flow. For decades chipmakers made the strips shorter, so the transistors switched faster and more of them fit on a chip. But once the strip gets short enough, current leaks through even when the gate is off, and the chip burns power doing nothing. The newest kind of transistor, called gate-all-around, cuts this leak by wrapping the gate around all four sides of the strip. A chipmaker needs two things to make the newest chips, and having one does not bring the other. The first is the know-how to build the newest switches, a long sequence of steps that have to run in exact order. In 2026 only TSMC, Intel and Samsung had it. The second is factories that cost tens of billions of dollars each, and in 2022 most of the world's capacity for the most advanced logic chips was in Taiwan. China's SMIC runs factories, but without EUV machines or the know-how for the newest switches, its most advanced chips are 7 nm-class. IBM developed 2 nm technology, but it yields no chips until Rapidus finishes a factory to run it, with mass production planned for 2027. No chip company sells enough of one product to fill such a factory on its own, so chip companies share a handful of them, run by foundries that make chips for other companies. TSMC runs the biggest, and all six of its largest plants are in Taiwan.

In short

Since September 2024 the United States has required a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls[1]. China has announced $43 billion to develop its own extreme ultraviolet (EUV) scanners, but in September 2025 the Center for Strategic and International Studies judged that EUV remains firmly out of reach[2].

SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027[3]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities[4]. SIA and BCG projected in May 2024 that the US share of the world's logic capacity below 10 nm would rise from almost none in 2022 to 28 percent by 2032[5].

Nvidia's Blackwell GPU holds 208 billion switches on two dies, each die the largest a lithography machine can print in one shot[6]. Every switch has to turn on fully, let almost no current through when off, and keep doing it for years. Blackwell is made by TSMC, on a custom process called 4NP[6]. TSMC calls a plant a GIGAFAB once it can run more than 100,000 wafers a month, each a 300 mm disc, twelve inches across[7]. It operates six of them, and every one is in Taiwan[8].

Chokepoint card Foundries and Fabs, stage 8 of 14

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Rapidus, a newcomer, said in July 2025 that it would begin 2 nm mass production in 2027[9].

Price or market size

$35-43 billion

over ten years to own a new 2 nm-class fab completed in 2026, BCG's 2023 estimate[10]. A 300 mm wafer on TSMC's 4NP process cost $17,000, Epoch AI estimated in December 2025[11].

1 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

2 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

4 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

6 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

7 GIGAFAB® Facilities. TSMC, accessed 9 September 2026. tsmc.com

8 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

9 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

10 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

11 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

Who leads	TW	**TSMC**	N2, its first nanosheet node, in volume production since 4Q25[12]
	US	**Intel**	18A in high-volume production in the United States, as of 2026[13]
	KR	**Samsung Foundry**	3 nm gate-all-around process in initial production since June 2022[14]
	CN	**SMIC**	$9.327bn revenue in 2025, with its lines 93.5% full[15]
	JP	**Rapidus**	267.6 billion yen raised in February 2026 to reach 2 nm mass production by 2027[16]
Where it is made	TW	**Taiwan**	TSMC's six gigafabs, with N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung, and UMC
	KR	**South Korea**	Samsung Foundry, Hwaseong and Pyeongtaek
	US	**United States**	Intel 18A at Fab 52 in Arizona, TSMC Arizona, Samsung Taylor and GlobalFoundries
	JP	**Japan**	Rapidus Chitose, Hokkaido, and TSMC Kumamoto
	CN	**China**	SMIC Shanghai and Beijing 300 mm lines, and Hua Hong

Why substitution is slow

Since September 2024 the United States has required a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls[1]. In March 2026 the Semiconductor Industry Association (SIA) put the cost to build and equip a leading-edge fab at as much as $20 to $25 billion[17]. SemiAnalysis judged in December 2025 that finding the highly skilled engineers a foundry needs remained a major challenge for TSMC in Arizona[18].

Where China stands

China's SMIC made its most advanced chips on N+3, a 7 nm-class process built without extreme ultraviolet scanners, SemiAnalysis found in June 2026[19]. SMIC cannot buy those scanners, so it prints its tightest layers in several passes on older deep-ultraviolet machines[19]. China will keep the world's largest installed fab capacity in 2026 and 2027, SEMI projected in its third-quarter 2026 forecast[3].

Where the US stands

Intel is the only chipmaker that both develops and manufactures leading-edge logic chips in the United States, by its own account in August 2025[20]. The United States had almost no logic capacity below 10 nm in 2022, and SIA and BCG projected in May 2024 that its share would reach 28 percent by 2032[5]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities[4]. The US government agreed in August 2025 to buy a passive 9.9 percent stake in Intel, with no board seat[20].

1 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

4 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

12 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

13 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

14 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

15 SMIC Announces 2025 Annual Results. SMIC, 26 March 2026. smics.com

16 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

17 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

18 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. newsletter.semianalysis.com

19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

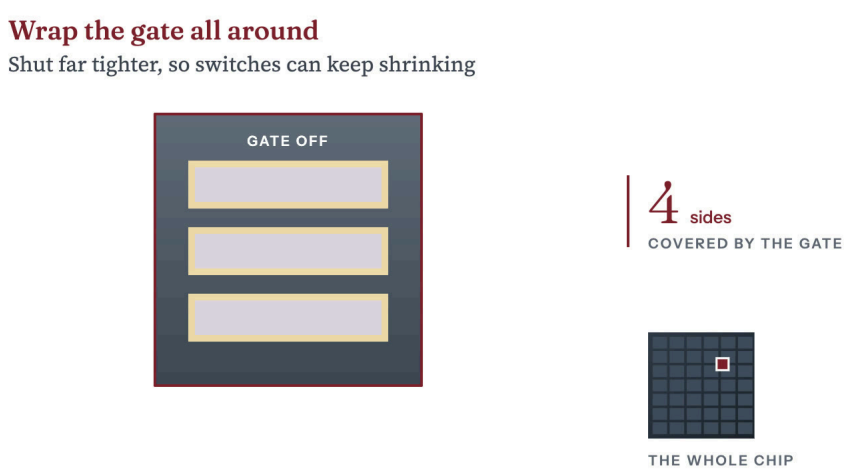
20 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

01

How it works

Figure 08.1 · Explainer, 23 seconds

Why the gate wraps around the channel



1. **Inside a chip.** An AI chip does all its work with transistors, tiny switches built into the silicon. One Nvidia Blackwell GPU holds 208 billion of them. Making the switches is the front end of the factory, and wiring them together comes after.
2. **On.** Each one is a switch. Seen end on, current flows through a strip of silicon, the channel, and the gate sits on top.
3. **Off, leaking.** Switching off, the gate squeezes the flow shut, but only near itself. In a tiny transistor, current still leaks through the part it cannot reach.
4. **Fin.** Stand the channel on edge as a fin, and the gate covers three sides. Much less gets through.
5. **All around.** Split it into thin sheets and wrap the gate around every side, and it shuts far tighter. That lets the switches keep shrinking, so each new chip can pack in more of them.

Note Simplified. The channel is seen end on, cut across the gate, with current drawn as points of light. Not to scale.

Current leaks in two main ways, straight through the gate insulator and along the channel from end to end while the gate is off. The second leak grows as the channel gets shorter. Three fixes deal with the two leaks:

- **Better insulator.** A thicker hafnium-based film, paired with a metal gate, lets fewer electrons slip straight through yet controls the channel as tightly as before. Intel replaced the silicon dioxide insulator with it at 45 nm in 2007[21]. It said the new film cut leakage through the insulator more than tenfold[21].

21 Intel's Transistor Technology Breakthrough Represents Biggest Change to... Intel, 27 January 2007. intel.com

- **More gate.** A planar transistor lays the gate on one side of a flat channel, and as the channel shortens the gate loses control of it. A FinFET stands the channel on edge as a fin, so the gate covers three sides, and gate-all-around wraps it around all four.
- **Thinner channel.** The fab thins the silicon until no current can flow where the gate cannot reach it.

02

How a transistor is built

Making the switches is called the front end of line. It builds them in a fixed order, starting with the channel. In a FinFET, the older design, the fab etches the silicon into thin fins that stand up from the wafer, and the gate goes over each one.

In a gate-all-around transistor, the fab instead grows alternating layers of silicon and silicon-germanium and cuts the stack into narrow bars[22]. It lays a placeholder, called a dummy gate, across each bar where the real gate will go[23]. On either side of the placeholder it cuts deep trenches where the source and drain, the two ends of the channel, will go[24]. Through those trenches it etches the silicon-germanium layers back a little and fills the gaps with an insulator, the inner spacer, which keeps the gate apart from the source and drain[22]. Then it grows the source and drain as crystal inside the trenches, a step called epitaxy[24].

Only then does the fab remove the placeholder and etch away the silicon-germanium, leaving thin silicon sheets that hang between the source and drain like shelves[22]. The inner spacer shields the source and drain from that etch[23]. The fab coats every sheet with an insulating film a few atoms thick, so the gate can control the channel without touching it. Gate metal then fills the gaps above, below and beside each sheet, covering all four sides[25]. Last come the contacts, small metal plugs down to the source, the drain and the gate, ready for the wiring layers above.

The steps have to run in exact order, because later layers cover earlier ones. Once gate metal surrounds a sheet, the insulating film can no longer go between the sheet and the metal. Each step is a separate deposition, etch or implant. An implant step, called ion implantation, fires atoms of boron or phosphorus into chosen patches of silicon so they conduct the way the design needs.

22 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com

23 Development of SiGe Indentation Process Control for Gate-All-Around FET... ArXiv, 12 January 2022. arxiv.org

24 Applied Materials Unveils Next-Gen Chipmaking Products to Supercharge AI... Applied Materials, 7 October 2025. ir.appliedmaterials.com

25 Applied Materials Technologies Enable 2D Scaling with EUV and 3D... Applied Materials, 21 April 2022. ir.appliedmaterials.com

03

The newest designs

Gate-all-around nanosheets

A nanosheet transistor lays the channel flat again, stacks two to four thin sheets and wraps gate metal around each one. Each fin in a FinFET adds a fixed amount of current, so a designer who needs more current has to add a whole fin. Nanosheets can be drawn at different widths, so current can be set in finer steps.

TSMC's N2, its first nanosheet node, entered volume production in the fourth quarter of 2025[12]. TSMC rates the follow-on N2P at 18 percent more speed at the same power and 36 percent less power at the same speed than its earlier N3E[26]. It gives N2P 1.2 times the logic density of N3E and 1.15 times its chip density, but its technology page does not define either measure[26]. These numbers are TSMC's own claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure that can be compared with a rival's.

Intel and Samsung also make gate-all-around transistors. Intel's version, called RibbonFET, is built into its 18A process, which Intel says is in high-volume production in the United States[13]. Samsung got there first, starting initial production of its 3 nm gate-all-around process in June 2022[14]. But SemiAnalysis reported in February 2023 that, because of yield problems, high-volume chips on Samsung's 3 nm gate-all-around technology were expected only in 2024[27].

SemiAnalysis reported in 2024 that the SRAM cell, which holds one bit in the fast memory on a logic die, had barely shrunk since the 5 nm node[28]. TSMC said in 2025 that its 2 nm SRAM uses a 0.021 square-micrometer cell and gets 1.1 times the density of its previous node by tuning design and process together[29]. See Memory and HBM.

Backside power delivery

In a conventional chip, power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. With backside power, the fab grinds the wafer thin and builds a second network for power underneath it.

12 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

13 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

14 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

26 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com

27 The Future of the Transistor. SemiAnalysis, 21 February 2023. newsletter.semianalysis.com

28 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com

29 A 38.1Mb/mm2 SRAM in a 2nm-CMOS-Nanosheet Technology for High-Density and... TSMC, 2025. research.tsmc.com

Intel's version, PowerVia, is part of 18A, and Intel calls it an industry first[13]. It cut routed area, the space a block of circuits takes once its wiring is laid out, by 11 percent and dynamic voltage droop tenfold[30]. Droop is the sag in supply voltage when a block of logic switches on at once. TSMC's Super Power Rail connects the backside network directly to each transistor's source and drain[28]. Against N2P, TSMC rates A16, which uses Super Power Rail, at 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density[26].

TSMC plans to start A14 production in 2028, with more than 20 percent higher logic density than N2[31]. It said in April 2026 that two more processes, A13 and A12, are scheduled to enter production in 2029[32].

Figure 08.2

TSMC plans four logic nodes beyond N2 for 2028 and 2029

TSMC's logic roadmap beyond N2, from its own announcements

- 2028
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028
  A14, the base node for A13 and A12
- 2029
  A13, a direct shrink of A14 for 6% area savings
- 2029
  A12, the A14 platform with Super Power Rail backside power

Source TSMC, 2026 North America Technology Symposium [32]

CFET and what comes after

A basic logic circuit needs two transistors that switch on opposite signals, and in production chips the two sit side by side. Complementary field-effect transistors, or CFETs, stack the two on top of each other[33]. The research institute imec sees them as an attractive design for logic nodes beyond 1 nm[33].

13 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com
26 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com
28 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com
30 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com
31 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com
32 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
33 CFET (complementary FET). Imec, 16 June 2022. imec-int.com

04

The back end of line

Wiring the transistors together is called the back end of line, and it has become as hard to make as the transistor. Copper, the usual wiring metal, seeps into the insulator around it, so every wire needs a barrier, a liner and a cap[34]. As the most critical wiring levels drop below a 20 nm pitch, those extra layers take a large share of the wire's width[34]. Pitch is the distance from the center of one wire to the center of the next. Two metals are moving into those narrow levels.

- **Ruthenium** shows promise for use without a barrier, which would keep the wire's resistance low[34]. imec has demonstrated ruthenium lines at a 16 nm pitch[35].
- **Molybdenum** in a thin film has up to 30 percent lower resistivity than tungsten, the metal it is replacing in contacts[36]. It needs no barrier either, so chipmakers are putting it into some contacts and local wiring layers[36].

05

The foundry and the fab

A foundry makes chips designed by other companies, running all their designs on the same tools. TSMC and UMC, both based in Taiwan, make chips only for other companies. TSMC, founded in 1987, pioneered this pure-play model[37].

Inside a fab, a chip is built up in layers, and a similar loop of steps repeats for each one. Wafers travel in batches, called lots, inside a sealed pod called a FOUP, short for front-opening unified pod. Entegris, a US maker of these pods, offers a standard version that holds 25 wafers[38]. An overhead track carries the pod from tool to tool.

At each tool a robot lifts one wafer out and the tool does its one job, such as depositing a film, printing a pattern, etching it, cleaning it or measuring it. Then the robot puts the wafer back, and the pod moves on to the next tool, often across the building. When a layer is finished, the loop starts again with the next one.

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor

34 Semi-damascene metallization. Imec, 31 October 2024. imec-int.com

35 16nm Ru lines using semi-damascene integration approach. Imec, 3 June 2025. imec-int.com

36 Molybdenum's Role in Ultra-Fast Computing: The Metal Behind the Speed. Entegris, 18 May 2026. blog.entegris.com

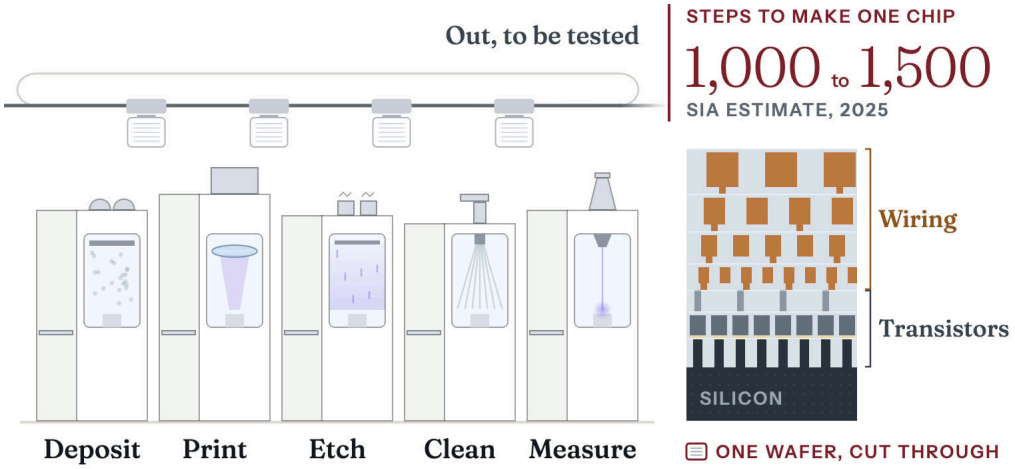
37 TSMC, company profile page. TSMC, accessed 9 September 2026. tsmc.com

38 A300 FOUPs: Front opening unified pod platform with super microenvironment... Entegris, 2021. entegris.com

Industry Association said in 2025[39]. In 2021 the association put the time to make a finished wafer at about 12 weeks on average, and up to 14 to 20 weeks for advanced processes[40]. The pod keeps the wafer clean for all of that time, because the wafer never meets the room's air.

Figure 08.3 · Explainer, 20 seconds

How a fab builds a chip, layer by layer



1. **The fab.** Sealed pods carry wafers from tool to tool on a track near the ceiling.
2. **Deposit.** The first tool lays a thin film on the wafer.
3. **Print.** The next prints a pattern on it with light.
4. **Etch.** An etch tool cuts the pattern into the film.
5. **Clean, measure.** The wafer is cleaned, then measured to check the layer came out right.
6. **Next layer.** Then the loop starts again for the next layer. The transistors come first, then the wiring above them.
7. **Finished.** Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Note Simplified. A real fab has many tools of each kind, spread across the building, and a chip has far more layers than the seven shown. The layers are drawn far thicker than they are.

Each tool is set by a recipe, the list of gases, temperatures and times for one step. A foundry offers every customer the same list of recipes, so every design in the building can run on the same shared tools. The tools run day and night, because an idle tool earns nothing. The finished wafer is tested (see Test and assembly), then cut and packaged.

39 Testimony of David Isaacs, Semiconductor Industry Association, hearing on… U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

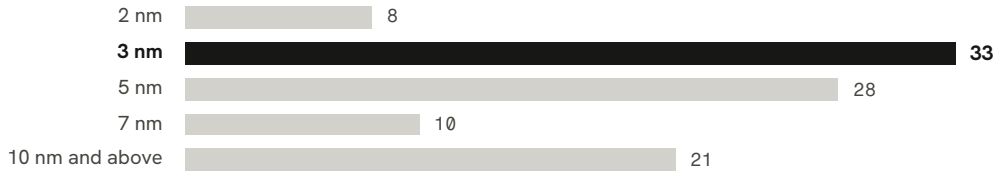
40 Chipmakers Are Ramping Up Production to Address Semiconductor Shortage. Here's… Semiconductor Industry Association, 26 February 2021. semiconductors.org

Chips at 7 nm and below, meaning the newest processes, were 79 percent of TSMC's wafer revenue in the third quarter of 2026[41]. That share split into 33 percent at 3 nm, 28 at 5 nm, 10 at 7 nm and 8 at 2 nm[41]. TSMC's revenue for the quarter was $46.72 billion, at a 66.4 percent gross margin[41].

Figure 08.4

3 nm led TSMC's wafer revenue in third-quarter 2026, at 33 percent

TSMC wafer revenue by node, 3Q26 % of wafer revenue



Source TSMC 3Q26 results, filed with the SEC [41]

Intel and Samsung design and sell their own chips and rent out capacity too. Renting out capacity is harder for them, because customers dislike handing their designs to a competitor and each company's own products take its best capacity. Intel Foundry reported $5.8 billion of segment revenue in the second quarter of 2026 and an operating loss of $2.089 billion[42]. Only $293 million of that revenue came from outside customers[43].

06

Capacity and cost

A new leading-edge fab costs tens of billions of dollars. In 2023 the Boston Consulting Group (BCG) estimated that a new fab for chips at 2 nm and below, completed in 2026, would cost $35 to $43 billion to build, equip and run over ten years[10]. Building and equipping alone takes as much as $20 to $25 billion of capital for a leading-edge fab, the Semiconductor Industry Association (SIA) told the US Senate in March 2026[17].

Most of TSMC's capacity is in its six GIGAFABs. Together, the plants TSMC and its subsidiaries run had capacity for more than 17 million wafers a year in 2025, counted as if each were the standard 12-inch size[8]. The six largest plants had a combined capacity of more than 13

8 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

10 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

17 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

41 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

42 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com

43 INTEL CORP, Form 10-Q quarterly report for the period ended 2026-06-27 (10-Q). U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

million[7]. The other plants are five smaller fabs in Taiwan, two in China, two in the United States and one in Japan[8]. In July 2026 TSMC announced another $100 billion for four more fabs at 2 nm or below, taking its planned investment in Arizona to $265 billion across 12 facilities[4].

In 2025 the biggest AI chip designers used most of the world's supply of two inputs but only a small share of its advanced chipmaking. Epoch AI estimated that the four largest used around 90 percent of the world's CoWoS packaging capacity and HBM memory supply[44]. CoWoS is TSMC's packaging for AI chips, and HBM is the stacked memory that sits beside them. The same four used only 12 percent of advanced logic production[44]. See Advanced packaging.

AI accelerators have usually trailed the leading edge. A die as large as theirs is more likely to hold a defect[45]. So the share of dies that work, called the yield, gets high enough only once a process has run long enough to cut its defect rate.

Nvidia's Blackwell is made on an older TSMC process than AMD's MI455X. Epoch AI estimated in December 2025 that a 300 mm wafer on Blackwell's 4NP process costs $17,000[11]. AMD's MI455X, launched on 23 July 2026, is made on TSMC's 2 nm and 3 nm processes[46]. It spreads its compute across eight separate dies[47].

07

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme ultraviolet (EUV) lithography. China's SMIC builds 7 nm-class nodes without it[19]. SemiAnalysis reported in September 2026 that Intel's Panther Lake, Samsung's Exynos 2600 and Apple's A20 Pro, which is made on TSMC's N2, all use gate-all-around transistors[48].

In August 2025 Intel called itself the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[20]. That month the US government agreed to buy 433.3 million newly issued Intel shares at $20.47 each, or $8.9 billion in all, for a 9.9 percent stake[20]. The stake is passive, with no board seat[20].

4 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

7 GIGAFAB® Facilities. TSMC, accessed 9 September 2026. tsmc.com

8 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

11 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

20 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

44 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

45 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

46 AMD Instinct™ MI455X GPUs. AMD, accessed 9 September 2026. amd.com

47 AMD Instinct MI455X GPU. AMD, 20 July 2026. amd.com

48 Intel Panther Lake Teardown. SemiAnalysis, 26 September 2026. newsletter.semianalysis.com

The deal also gave the government a five-year warrant to buy a further 5 percent of Intel at $20 a share[20]. It can use the warrant only if Intel's ownership of its foundry business falls below 51 percent[20].

In 2026 Intel began high-volume manufacturing of some of its Panther Lake processors using ASML's High-NA extreme ultraviolet scanners[42]. But it is building fewer fabs than it planned. In July 2025 Intel said it would no longer go ahead with planned projects in Germany and Poland and would slow construction in Ohio[49].

In December 2024 the US Commerce Department awarded Samsung up to $4.745 billion of CHIPS Act funding, to support more than $37 billion of planned investment in Central Texas[50]. That plan covers two new leading-edge logic fabs and a research fab in Taylor, and an expansion of Samsung's existing plant in Austin[50].

China's SMIC has the factories without the know-how for the newest transistors. Its revenue rose 16.2 percent to $9.327 billion in 2025, at a 21 percent gross margin[15]. Its lines ran 93.5 percent full, on monthly capacity above a million wafers counted in the smaller 8-inch size[15].

A node's name, such as 7 nm, does not measure any feature on the chip. SMIC's N+3, its third-generation 7 nm process, sets its narrowest wires 32.5 nm apart, tighter than the 36 nm in Intel's Panther Lake chips on 18A[19]. But the 18A process itself allows wires 32 nm apart[19]. SemiAnalysis called the comparison an incomplete, cherry-picked metric[19]. N+3 packs 113.4 million transistors per square millimeter, against 107.7 for TSMC's N6[19].

Two research groups disagree about where Huawei's Ascend AI chips are made. Georgetown University's Center for Security and Emerging Technology (CSET) reported in June 2024 that the Ascend 910B was believed to be made on SMIC's N+2 process[51]. But the research firm SemiAnalysis judged in April 2025 that most Ascend 910B and 910C chips were made on TSMC's 7 nm process[52]. It said Huawei got around the sanctions that barred it from TSMC by buying about $500 million of 7 nm wafers through another company, Sophgo[52].

IBM shows the opposite split. Its 2 nm technology yields no chips until Rapidus, a state-backed newcomer in Japan, finishes a factory to run it. The two firms agreed in December 2022 to develop that technology further for production at Rapidus's plant in Japan[53].

15 SMIC Announces 2025 Annual Results. SMIC, 26 March 2026. smics.com

19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

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50 Biden-Harris Administration Announces CHIPS Incentives Award with Samsung... NIST, 20 December 2024. nist.gov

51 Pushing the Limits: Huawei's AI Chip Tests U.S. Export Controls. Center for Security and Emerging Technology, 17 June 2024. cset.georgetown.edu

52 Huawei AI CloudMatrix 384. SemiAnalysis, 16 April 2025. newsletter.semianalysis.com

53 IBM and Rapidus Form Strategic Partnership to Build Advanced Semiconductor... Rapidus, 13 December 2022. rapidus.inc

In July 2025 Rapidus said it had started making 2 nm prototypes at the plant, in Chitose, Hokkaido, and would begin mass production in 2027[9]. In February 2026 it raised 267.6 billion yen from the Japanese government and 32 companies to reach 2 nm mass production by 2027[16]. By June 2026 Rapidus had raised 424.95 billion yen in equity[54].

08

The chokepoint

A firm that wants to make the newest chips has to develop its own process, because TSMC, Intel and Samsung do not sell their leading-edge process flows. A process flow sets the steps, their order and the tools for each. In September 2024 the United States began requiring a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls[1].

China has money for chipmaking but cannot buy EUV scanners. It launched its third National Integrated Circuit Fund in May 2024 with 344 billion yuan, about $47 billion[55]. China's 7 nm chips have been made on deep-ultraviolet machines, a generation behind EUV[2]. China had stockpiled those machines from ASML, the Dutch scanner maker, before export restrictions took effect[2]. SMIC gets fine features out of them by building its tightest layers in several patterning steps[19].

In July 2026 Reuters reported that a Shanghai state firm had begun making home-grown immersion deep-ultraviolet scanners[56]. Reuters's own source said the firm's machine still needed testing and was far from matching ASML's models[56].

Domestic EUV is further off. China has announced $43 billion to develop it[2]. In September 2025 the Center for Strategic and International Studies (CSIS) judged that EUV remains firmly out of reach despite massive state investment[2]. See Lithography.

The factories are concentrated too. In 2022 South Korea and Taiwan held almost all of the world's logic capacity below 10 nm, and the United States had almost none[57]. Taiwan alone held 69 percent[5]. In

1 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

2 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

9 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

16 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

54 Rapidus Completes 150 Billion Yen Funding Round from Japan Government. Rapidus, 5 June 2026. rapidus.inc

55 China launches $47bn chip fund to counter U.S. restrictions. Nikkei Asia, 27 May 2024. asia.nikkei.com

56 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

57 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 13. semiconductors.org

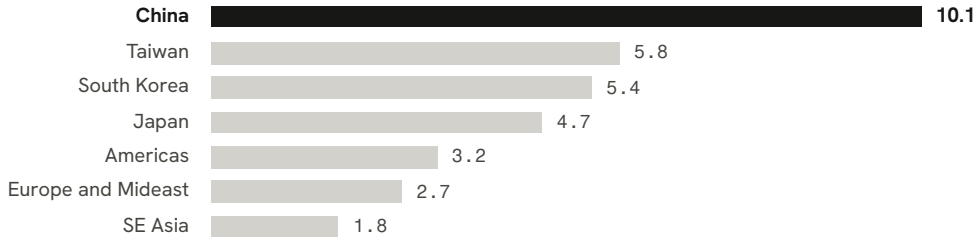
May 2024 SIA and BCG projected that the US share of that capacity would reach 28 percent by 2032[5].

China leads the world in capacity for logic chips made at 28 nm or older, but little of its capacity is advanced[58]. Most of the world's wafers come off such older lines, which make power devices, analog chips and microcontrollers. Across all nodes, old and new, the industry association SEMI forecast in June 2024 that China would have installed capacity of 10.1 million wafers a month in 2025, the most of any region[59]. It forecast 5.8 million for Taiwan, out of a world total of 33.7 million, all counted in the 8-inch size[59]. In the third-quarter 2026 update of its World Fab Forecast, SEMI projected that China would keep the largest installed capacity in 2026 and 2027[3].

Figure 08.5

China led SEMI's 2025 capacity forecast at 10.1 million 8-inch-equivalent wafers monthly

SEMI's June 2024 forecast of installed wafer fab capacity by region in 2025, all nodes million 8-inch-equivalent wafers per month



Source SEMI World Fab Forecast, June 2024 [59]

Chip production in Taiwan faces three risks:

- **Seismic.** Taiwan sits on an active plate boundary, and after an earthquake in January 2025 TSMC had to scrap some wafers in process[60].
- **Power.** TSMC says power outages, dips or surges caused by problems at its electricity supplier have at times interrupted its operations[61].
- **Political.** In December 2025 SemiAnalysis put Taiwan's share of the world's advanced-node capacity at over 90 percent, though it did not say which node sizes count as advanced[18]. So a blockade or

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

18 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. newsletter.semianalysis.com

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59 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in... SEMI, 18 June 2024. semi.org

60 TSMC January 2025 Revenue Report and the Statement on the Impact of Earthquake. TSMC, 10 February 2025. pr.tsmc.com

61 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 20-F annual report for the... U.S. Securities and Exchange Commission, 16 April 2026. sec.gov

conflict would cut most of that output off from the world market at once. See Who Controls the Chain.

Building capacity outside Taiwan is slow, and TSMC has yet to build a gigafab anywhere else[8]. SemiAnalysis wrote in December 2025 that a foundry needs highly skilled engineers to get the most working chips from each wafer[18]. It judged that finding enough workers remained a major challenge for TSMC in Arizona[18].

Five questions to answer from memory

1. What two things does a chipmaker need to make the newest chips? see How it works
2. Why do the newest transistors wrap the gate around the channel? see How it works
3. What is a foundry? see The foundry and the fab
4. Where are TSMC's largest plants? see Capacity and cost
5. How does China's SMIC make advanced chips without an EUV machine? see The chokepoint

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Five things to remember

- The newest chips need both the know-how to build the newest transistors and factories that cost tens of billions of dollars, and in 2026 only TSMC, Intel and Samsung had both.
- The newest transistors wrap the gate around all four sides of the channel to cut the current that leaks when the switch is off.
- A foundry makes chips for other companies, because no chip company sells enough of one product to fill a fab alone.
- All six of TSMC's largest plants are in Taiwan, which held 69 percent of the world's logic capacity below 10 nm in 2022.
- China's SMIC makes 7 nm-class chips without EUV by building its tightest layers in several patterning steps.

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